XIAMEN UNIVERSITY MALAYSIA
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EEE312 Week2

Lab Practice

Instructions:

- Write each exercise in to codeblocks in separate C file
- Run each program, understand the results

Exercise

```
// Write a C code to find the determinant of an N×Nmatrix by performing
the row operations.

// Sample code for Lab practice
// determinant.c - find the determinant for an NxN matrix
#include <stdio.h>
#define N 4 // size of the square matrix is 4 x 4

void rowoperation(double [][N]); // function to perform row operation
double determinant(double [][N]); // function to calculate determinant of matrix
void printMatrix(double [][N]); // function to print matrix
```

Function printMatrix()

```
void printMatrix(double A[][N])
{
    for(int i=0; i<N; i++) // for every row index from i=0 to N-1
        for(int j=0; j<N; j++) // for every column index from j=0 to N-1
        {
            printf("%.4f\t", A[i][j]); // print the element A[i][j] followed by a tab
            if(j==N-1) printf("\n"); // if reach the end of row, print a newline
        }
    printf("\n");// print a newline character after the printing the matrix
}
```

C statements to perform row operation

```
m = A[i][k]/A[k][k]; // A[k][k] is the pivot element
for (int j=0; j<N; j++) // For row i , perform element wise
    // operation from j=0 to N 1. j is the column index
    A[i][j] -= m*A[k][j];
//At row i , perform i ][ R[ i ][ m*R[k][j]
```

C statements to perform row operation (continued)

```
// Next, we repeat the previous code for row i= k+1 to N-1.
// k= row index where the pivot element is located.
```

```
double m;
for (int k=0; k<N 1; k++) // do for row k=0 to k=N-2;
    // k is the row index where the pivot element is located
    // in this part, the codes from the previous step is placed.
```

Function rowoperation()

```
void rowoperation(double A[][N])
{
    // row operations
    double m;
    for(int k=0; k<N-1; k++) // do for row k=0 to k=N-2; k is the row index where the pivot element is located
    {
        for(int i=k+1; i<N; i++) // do for row below k from k+1 to N-1; i is the row index where there is NO pivot element
        {
            m = A[i][k]/A[k][k]; // A[k][k] is the pivot element
            for(int j=0; j<N; j++) // For row i, perform element wise operation from j=0 to N-1. j is the column index
                A[i][j] -= m*A[k][j]; // At row i, perform R[i][j]<-R[i][j]-m*R[k][j]

        }
    }
}
```

Function determinant()

```
double determinant(double A[][N])
{
    double det=1.0; // store the result of the determinant calculation
    for(int i = 0; i<N; i++) // for row i = 0 to N-1 of matrix A
        det *= A[i][i]; // multiply the diagonal element A[i][i]
    return det; // return the result of the determinant
}
```

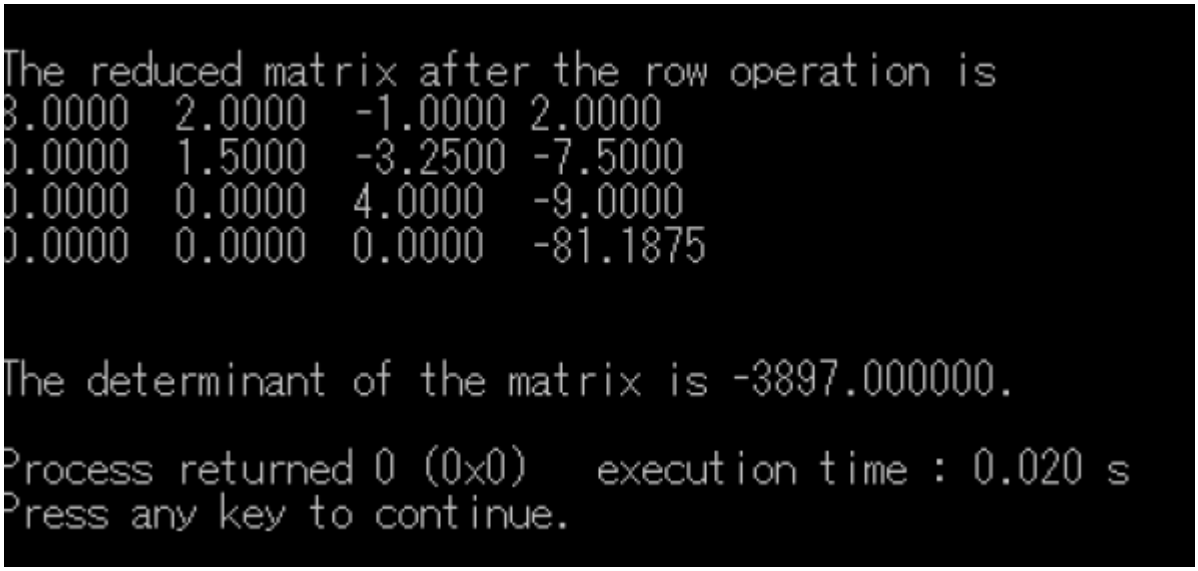
Function main()

```
int main()
{

    double det = 0; // store the value of the determinant
    double M[N][N] = {{8.0,2,-1,2},{-2,1,-3,-8},{2,-1,7,-1},{1,-8,1,-2}};// declare and initialize matrix M

    rowoperation(M); // call the function rowoperation() to perform row operation on matrix M
    printf("\nThe reduced matrix after the row operation is \n");
    printMatrix(M); // call the function printMatrix() to print out the triangular matrix M
    det = determinant(M); // call the function determinant to calculate the determinant of matrix M
    printf("\nThe determinant of the matrix is %f.\n", det); // print the result of the determinant on screen
    return 0;
}
```

Output


The reduced matrix after the row operation is
3.0000 2.0000 -1.0000 2.0000
0.0000 1.5000 -3.2500 -7.5000
0.0000 0.0000 4.0000 -9.0000
0.0000 0.0000 0.0000 -81.1875
The determinant of the matrix is -3897.000000.
Process returned 0 (0x0) execution time : 0.020 s
Press any key to continue.